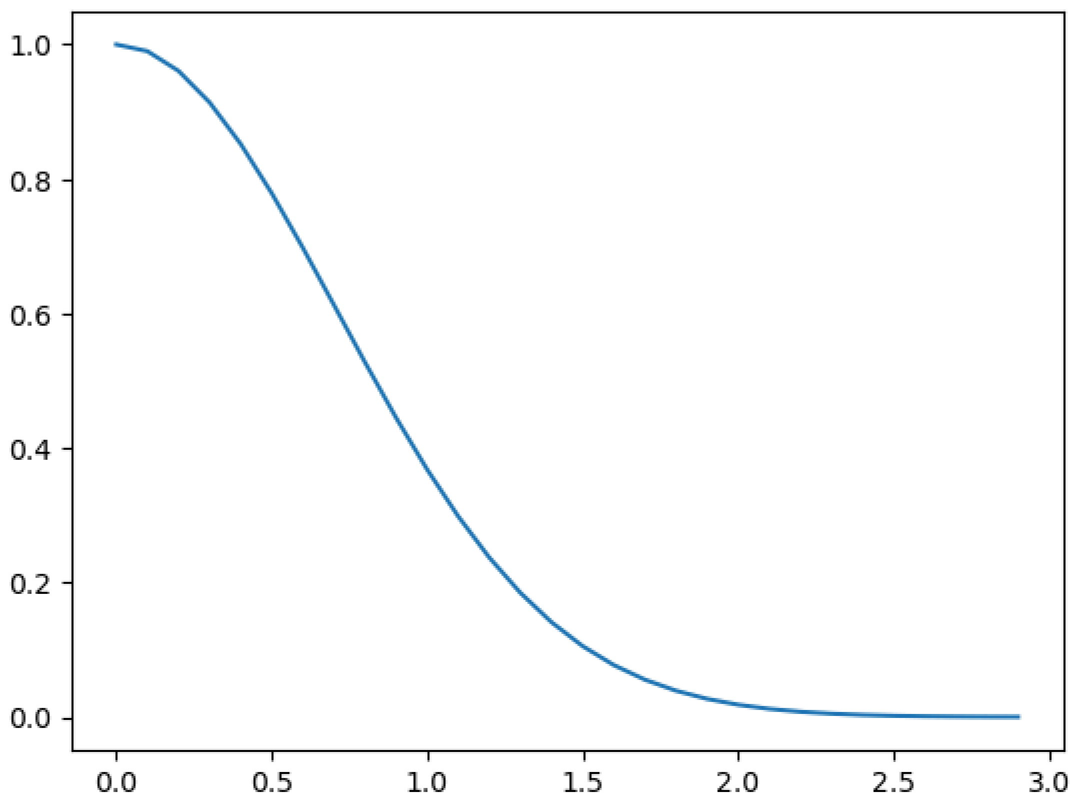
5.03

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [9]:
```
def g(t):
    return(np.exp(-(t**2)))
t = np.arange(0,3,0.1)

fig, ax = plt.subplots()
ax.plot(t, g(t))
```

Out[9]:
```
[<matplotlib.lines.Line2D at 0x70597b97ee70>]
```



In [10]:
```
def simp1(func, a, b, steps):
    if steps%2!=0:
        steps = steps + 1
        print('i added one step')
    dx = (b-a)/steps
    x = np.linspace(a, b, steps+1)
    y = func(x)
    s = dx/3*(y[0]+y[-1])
    s = s + dx/3*(4*np.sum(y[1:steps:2]) + 2*np.sum(y[2:steps-1:2]))
    return(s)
```

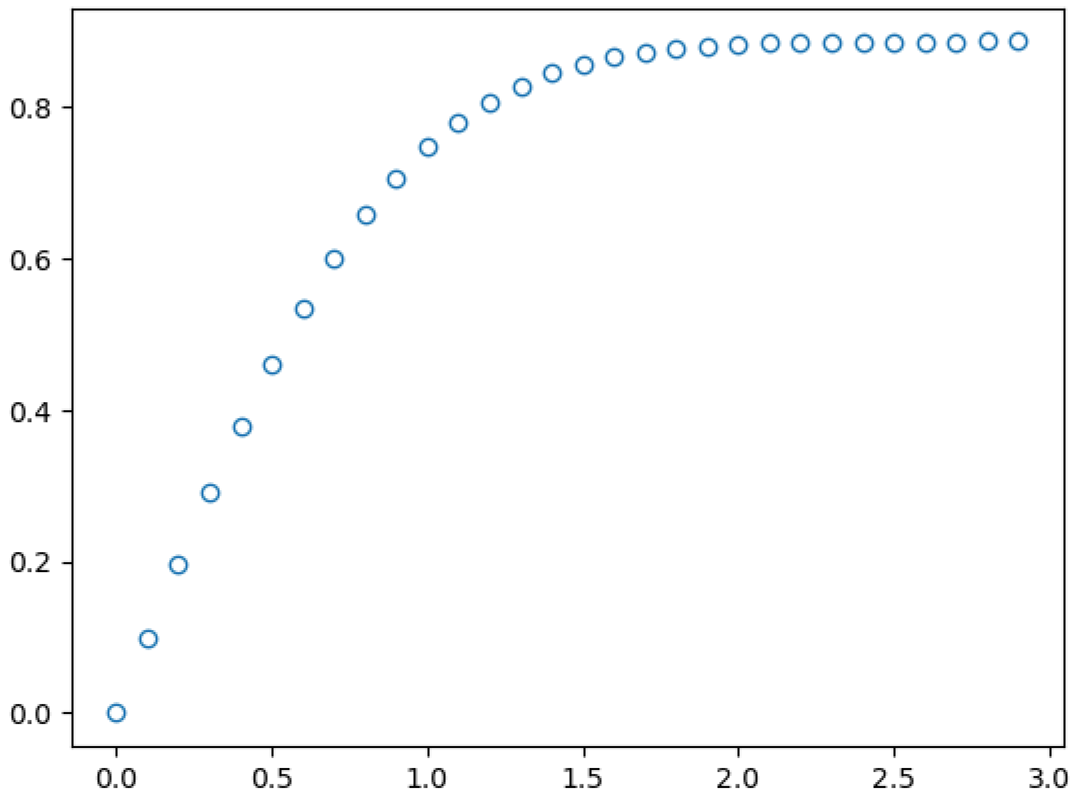
```python
def f(x):
    def g(t):
        return(np.exp(-(t**2)))
    integral = 0.0
    integral = simp1(g, 0 , x, 1000)
    return(integral)

x = np.arange(0,3,0.1)
y = []
for i in x:
    y.append(f(i))

fig, ax = plt.subplots()

ax.plot(x, y, 'o', mfc='none')
```

Out[10]: [<matplotlib.lines.Line2D at 0x70597b810d40>]



In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [5]:
```python
x = np.arange(0,3.1,0.1)
```

In [6]:
```
x
```

Out[6]:
```
array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. , 1.1, 1.2,
       1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2. , 2.1, 2.2, 2.3, 2.4, 2.5,
       2.6, 2.7, 2.8, 2.9, 3. ])
```

In [ ]:
```

```